

Directions for Use

Please Read Carefully!

Nimodipine HEXAL® 30 mg Film coated Tablet

Content

Each **Nimodipine HEXAL® 30 mg Film-coated Tablet** contains nimodipine 30 mg.

Description

Nimodipine HEXAL® 30 mg Film-coated Tablet: Yellow to orange, round, biconvex film-coated tablets without score.

Properties

Nimodipine is a calcium antagonist belonging to the 1, 4-dihydropyridine group with preferential activity on cerebral vessels. In animal experiments, nimodipine binds to Ca^{2+} -channels of the L-type with high affinity and selectivity and blocks the transmembranous Ca^{2+} -influx. It is supposed that nimodipine might improve stability and efficiency in pathological conditions associated with an increased influx of Ca^{2+} -ions into nerve cells, e.g. in case of cerebral ischaemia

Nimodipine dilates the cerebral blood vessels in acute cerebral blood flow disturbances patients and increase cerebral perfusion. The increase in perfusion particularly in poorly perfused areas, by arterial dilatation is greater in previously damaged or underperfused brain regions than in healthy regions. Hence, the ischaemic neurological damage and lethality occurring in connection with subarachnoid haemorrhage were thus significantly reduced by nimodipine.

A selective blockage of Ca^{2+} -channels in certain cerebral areas, such as hippocampus and cortex, possibly explains the positive influence of nimodipine on learning and memory deficits observed in various animal models. The same molecular mechanism probably forms the basis of the cerebral vasodilative and perfusion-stimulating effect of nimodipine observed in animals and humans. It will be subject to prospective studies to find out in which way the molecular effects of nimodipine lead to an improvement in cerebral capacity dysfunctions which were ascertained by using neurophysiologic and clinical test models.

Pharmacokinetics

Absorption

The orally administered agent nimodipine is virtually completely absorbed. The unchanged agent and its early "first-pass" metabolic products can be identified in plasma 10-15 minutes after administration of the tablet. After multiple oral administration (30 mg 3 times daily), mean maximum plasma concentrations (C_{max}) of 12.3-17.5 ng/ml are measured in young patients and 26±10 ng/ml in elderly patients; they are reached after 0.6-1.6 hours (t_{max}). The maximum plasma concentration and the area under the curve (AUC) increase up to the highest tested dosage of 90 mg in proportion to the dose.

Protein binding and distribution

Nimodipine is bound to plasma proteins at 97-99 %. Nimodipine added to full blood distributes between erythrocytes and plasma in approx. equal parts. Nimodipine has proven in animal studies to cross the placental barrier and the transfer into mother's milk could be ascertained in the lactating rat. Both processes are probable to occur in human as well, although no data is available to date.

Nimodipine permeates the blood brain barrier well due to its high lipophilic properties. It can be identified in the cerebrospinal fluid after oral and i.v. administration at concentrations which are about 0.5 % of the measured plasma concentrations. These approximately correspond to the free concentration in plasma.

Metabolism, elimination and excretion

Nimodipine is eliminated metabolically mainly via dehydrogenation of the dihydropyridine ring and oxidative O-demethylation. Oxidative ester cleavage, hydroxylation of the 2- and 6-methyl group and glucuronisation as conjugation reaction are further important metabolic processes. The three primary metabolites occurring in plasma do not show any or only therapeutically insignificant residual activity.

Effects on liver enzymes by induction or inhibition are unknown. The metabolites are excreted in humans at approximately 50 % via the kidneys and at approximately 30 % via the bile.

The elimination kinetics are linear. The half-life for nimodipine is between 1.1 and 1.7 hours. The terminal half-life of 5-10 hours is not relevant for the determination of the dose interval.

Bioavailability

Due to extensive first pass metabolism in the liver (approximately 85-90%) the absolute bioavailability is 10-15%.

Indications

Prophylaxis and treatment of ischaemic neurological deficits due to cerebral vasospasm after subarachnoid haemorrhage.

Adverse Reactions

The frequencies of ADRs reported with nimodipine summarized in the tables below are based on clinical trials with nimodipine in the indication aSAH sorted by CIOMS III categories of frequency (placebo-controlled studies: nimodipine N = 703; placebo N = 692; uncontrolled studies: nimodipine N = 2496; status: 31 Aug 2005). Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Frequencies are defined as:

Very common	(≥ 1/10)
Common	(≥ 1/100 to < 1/10)
Uncommon	(≥ 1/1,000 to < 1/100)
Rare	(≥ 1/10,000 to < 1/1,000)
Very rare	(< 1/10,000)

System organ class (MedDRA)	Uncommon	Rare
Blood and the lymphatic system disorders	Thrombocytopenia	
Immune system disorders	Allergic reaction Rash	
Nervous system disorders	Headache	
Cardiac disorders	Tachycardia	Bradycardia
Vascular disorders	Hypotension Vasodilatation	
Gastrointestinal disorders	Nausea	Ileus
Hepatobiliary disorders		Transient increase in liver enzymes

Warning and Precautions

In very old multimorbid patients, in case of severely impaired kidney function (glomerular filtration rate < 20 ml/min), and in those with severely impaired cardiovascular function the need for treatment with nimodipine should be carefully considered and follow-up examination should be carried out regularly.

Although treatment with nimodipine has not been shown to be associated with increases in intracranial pressure, close monitoring is recommended in these cases or when the water content of the brain tissue is elevated (generalized cerebral edema).

Caution is required in patients with hypotension (systolic blood pressure lower than 100 mm Hg).

In patients with unstable angina or within the first 4 weeks after acute myocardial infarction, physicians should consider the potential risk (e.g. reduced coronary artery perfusion and myocardial ischemia) versus the benefit (e.g. improvement of brain perfusion).

Nimodipine is metabolized via the cytochrome P450 3A4 system. Medicinal products that are known to either inhibit or to induce this enzyme system may therefore alter the first pass or the clearance of nimodipine (see **Drug Interactions**).

Medicinal products, which are known inhibitors of the cytochrome P450 3A4 system and therefore may lead to increased plasma concentrations of nimodipine are, e.g.

- macrolide antibiotics (e.g. erythromycin)
- anti-HIV protease inhibitors (e.g. ritonavir)
- azole antimycotics (e.g. ketoconazole)

- the antidepressants nefazodone and fluoxetine
- quinupristin/dalfopristin
- cimetidine
- valproic acid.

Upon co-administration with these drugs, the blood pressure should be monitored and, if necessary, a reduction of the nimodipine dose should be considered.

Effects on the ability to drive and to use machines

In principle the ability to drive and use machines can be impaired in connection with the possible occurrence of dizziness.

Contraindications

Nimodipine must not be administered in case of hypersensitivity to the active substance or to any of the excipients.

The use of nimodipine in combination with rifampicin is contraindicated as efficacy of nimodipine could be significantly reduced when concomitantly administered with rifampicin (see **Drug Interactions**).

The concomitant use of oral nimodipine and the antiepileptic medicinal product phenobarbital, phenytoin or carbamazepine is contraindicated as efficacy of nimodipine could be significantly reduced (see **Drug Interactions**).

Severely disturbed liver function, particularly liver cirrhosis, may result in an increased bioavailability of nimodipine due to a decreased first pass capacity and a reduced metabolic clearance. Therefore, nimodipine must not be administered for treatment of ischemic neurological deficits to patients with severely impaired liver function (e.g. cirrhosis of the liver).

Drug Interactions

Medicinal products that affect nimodipine

Nimodipine is metabolised via the cytochrome P450 3A4 system located both in the intestinal mucosa and in the liver. Medicinal products that are known to either inhibit or to induce this enzyme system may therefore alter the first pass or clearance of nimodipine (see **Recommended Dosage, Patients with hepatic impairment**).

The extent as well as the duration of interactions should be taken into account when administering nimodipine together with the following drugs:

Rifampicin

From the experience with other calcium antagonists it has to be expected that rifampicin accelerates the metabolism of nimodipine due to enzyme induction. Thus, efficacy of nimodipine could be significantly reduced when concomitantly administered with rifampicin. The use of nimodipine in combination with rifampicin is therefore contraindicated (see **Contraindications**).

Cytochrome P450 3A4 system inducing anti-epileptic medicinal products, such as phenobarbital, phenytoin or carbamazepine

Previous chronic administration of the antiepileptic medicinal products phenobarbital, phenytoin or carbamazepine markedly reduces the bioavailability of orally administered nimodipine. Therefore, the concomitant use of oral nimodipine and these antiepileptic drugs is contraindicated (see **Contraindications**).

Upon co-administration with the following inhibitors of the cytochrome P450 3A4 system the blood pressure should be monitored and, if necessary, an adaptation in the nimodipine dose should be considered.

Macrolide antibiotics (e.g. erythromycin)

No interaction studies have been carried out between nimodipine and macrolide antibiotics. Certain macrolide antibiotics are known to inhibit the cytochrome P450 3A4 system and the potential for medicinal product interaction cannot be ruled out at this stage. Therefore, macrolide antibiotics should not be used in combination with nimodipine (see **Warning and Precautions**).

Azithromycin, although structurally related to the class of macrolide antibiotic is void of CYP3A4 inhibition.

Anti-HIV protease inhibitors (e.g. ritonavir)

No formal studies have been performed to investigate the potential interaction between nimodipine and anti-HIV protease inhibitors. Medicinal products of this class have been reported to be potent inhibitors of the cytochrome P450 3A4 system. Therefore, the potential for a marked and clinically relevant increase in nimodipine plasma concentrations upon co-administration with these protease inhibitors cannot be excluded (see **Warning and Precautions**).

Azole anti-mycotics (e.g. ketoconazole)

A formal interaction study investigating the potential of medicinal product interaction between nimodipine and ketoconazole has not been performed. Azole anti-mycotics are known to inhibit the cytochrome P450 3A4 system, and various interactions have been reported for other dihydropyridine calcium antagonists. Therefore, when administered together with oral nimodipine, a substantial increase in systemic bioavailability of nimodipine due to a decreased first-pass metabolism cannot be excluded (see **Warning and Precautions**).

Nefazodone

No formal studies have been performed to investigate the potential interaction between nimodipine and nefazodone. This antidepressant medicinal product has been reported to be a potent inhibitor of the cytochrome P450 3A4. Therefore, the potential for an increase in nimodipine plasma concentrations upon co-administration with nefazodone cannot be excluded (see **Warning and Precautions**).

Fluoxetine

The steady-state concomitant administration of nimodipine with the antidepressant fluoxetine led to about 50% higher nimodipine plasma concentrations. Fluoxetine exposure was markedly decreased, while its active metabolite norfluoxetine was not affected (see **Warning and Precautions**).

Quinupristin/dalfopristin

Based on experience with the calcium-antagonist nifedipine, co-administration of quinupristin/dalfopristin may lead to increased plasma concentrations of nimodipine (see **Warning and Precautions**).

Cimetidine

The simultaneous administration of the H2-antagonist cimetidine can lead to an increase in the plasma nimodipine concentration (see **Warning and Precautions**).

Valproic acid

The simultaneous administration of the anticonvulsant valproic acid can lead to an increase in the plasma nimodipine concentration (see **Warning and Precautions**).

Further medicinal product interaction

Nortriptyline

The steady-state concomitant administration of nimodipine and nortriptyline led to a slight decrease in nimodipine exposure with unaffected nortriptyline plasma concentrations.

Effects of nimodipine on other drugs

Blood pressure lowering medicinal products

Nimodipine may increase the blood pressure lowering effect of concomitantly applied anti-hypertensives, such as:

- diuretics
- β -blockers
- ACE inhibitors
- A1-antagonists
- other calcium antagonists
- α -adrenergic blocking agents
- PDE5 inhibitors
- α -methyl dopa.

However, if a combination of this type proves unavoidable particularly careful monitoring of the patient is necessary.

Zidovudine

In a monkey study simultaneous administration of anti-HIV drug zidovudine i.v. and nimodipine bolus i.v. resulted for zidovudine in significantly higher AUC, whereas the distribution volume and clearance were significantly reduced.

Medicinal product-food interactions

Grapefruit juice

Grapefruit juice inhibits the cytochrome P450 3A4 system. Administration of dihydropyridine calcium antagonists together with grapefruit juice thus results in elevated plasma concentrations and prolonged action of nimodipine due to a decreased first pass metabolism or reduced clearance.

As a consequence, the blood pressure lowering effect may be increased. After intake of grapefruit juice this effect may last for at least 4 days after the last ingestion of grapefruit juice. Ingestion of grapefruit / grapefruit juice is therefore to be avoided while taking nimodipine.

Pregnancy

There are no adequate and well controlled studies in pregnant women. If nimodipine is to be administered during pregnancy, the benefits and the potential risks must therefore be carefully weighted according to the severity of the clinical picture.

Lactation

Nimodipine and its metabolites have been shown to appear in human milk at concentrations of the same order of magnitude as corresponding maternal plasma concentrations. Nursing mothers are advised not to breastfeed their babies when taking the medicinal product.

Fertility

In single cases of in vitro fertilization calcium antagonists have been associated with reversible biochemical changes in the spermatozoa's head section that may result in impaired sperm function. The relevance of this finding in short-term treatment is unknown.

Recommended dosage

The recommended procedure is administration of nimodipine infusion solution for 5-14 days, followed by a daily dose of 6 x 2 **Nimodipine HEXAL ®30 mg Film-coated Tablet** (6 x 60 nimodipine).

Administration of **Nimodipine HEXAL ® 30 mg Film-coated Tablet** is recommended for about 7 days after the end of 5-14 day infusion therapy with nimodipine infusion solution. In general, the tablets should be swallowed in whole with a little liquid, independently of meal times. The interval between successive doses must not be less than 4h.

Prophylactic use:

Intravenous therapy should be started no later than 4 days after the haemorrhage and the continued during the period of maximum risk of vasospasm, i.e. up to 10-14 days after the subarachnoid haemorrhage. After the end of the infusion therapy it is advisable to continue with oral administration of 6 x 60mg nimodipine a day at 4 h interval for about a further 7 days.

Therapeutic administration:

If ischaemic neurological disturbances caused by vasospasm after subarachnoid haemorrhage are already present, treatment should be started as early as possible and be continued for at least 5 days up to a maximum of 14 days.

Thereafter oral administration of 6 x 60mg nimodipine/day at 4 h interval for 7 days is recommended.

Patients with hepatic impairment

Severely disturbed liver function, particularly liver cirrhosis, may result in an increased bioavailability of nimodipine due to a decreased first pass capacity and a reduced metabolic clearance. The effects and side effects, e.g. reduction in blood pressure, may be more pronounced in these patients.

In such cases the dose should be reduced or, if necessary, discontinuation of the treatment should be considered.

Paediatric population

Safety and efficacy of nimodipine in patients under 18 years of age have not been established.

Method and duration of administration

The tablets should be swallowed whole with some liquids irrespective of meal times. The interval between the intake times should not be less than 4 hours.

Symptoms and Treatment of Overdosage and Antidote

Symptoms of intoxication

Symptoms of acute overdose to be anticipated are marked lowering of the blood pressure, tachycardia or bradycardia, and gastrointestinal complaints and nausea.

Treatment of intoxication

In the event of acute overdose treatment with nimodipine must be discontinued immediately.

Emergency measures should be governed by the symptoms.

Gastric lavage with addition of charcoal should be considered as an emergency therapeutic measure. If there is a marked fall in blood pressure, dopamine or noradrenaline can be administered intravenously. Since no specific antidote is known, subsequent treatment for other side effects should be governed by the most prominent symptoms.

Presentation:

Box of 30, 60 & 120 film-coated tablets.

Shelf life: Please refer to the outer box label.

Do not store above 25 °C.

Keep out of reach of children.

Jauhi daripada kanak-kanak.

Keep medicines out of reach of children

Manufactured for:

HEXAL AG

Industriestraße 25

Holzkirchen, Germany

By:

Salutas Pharma GmbH

(A wholly owned subsidiary of HEXAL AG)

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